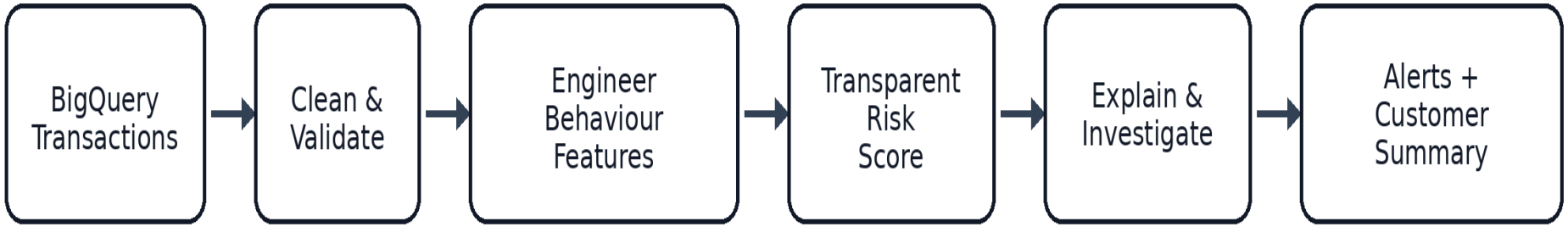
AML Transaction Monitoring, Risk Scoring & Investigation Assistant

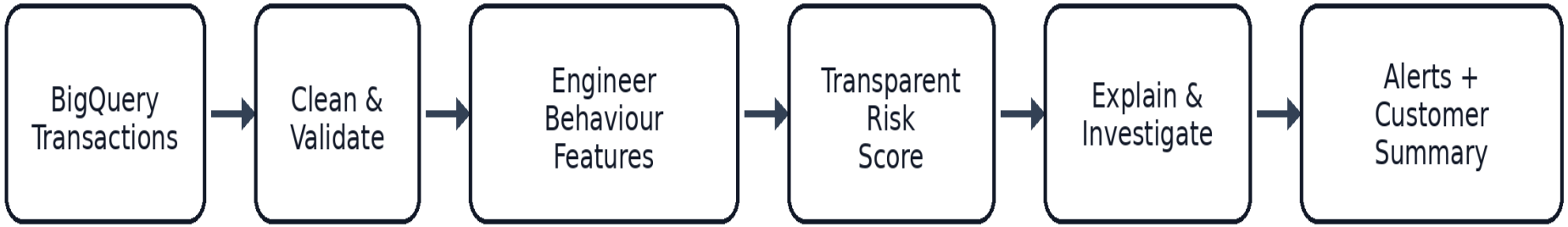
Simple-language concept guide based on the attached BigQuery and Python project



Interview revision focus: data engineering, AML analytics, explainable scoring, validation and investigator support.

Project at a glance

Business problem	Millions of transactions cannot be reviewed manually. The project prioritises unusual transactions and accounts for human investigation.
Data source	A BigQuery table containing sender, receiver, amount, currency, bank, payment format, time and an optional laundering label.
Core outputs	Transaction alert table, customer-level risk summary, plain-language risk reasons, recommended actions and draft case narratives.
Design principle	A transparent score is used so investigators can see why an alert was created instead of receiving a black-box prediction.
Human control	The system recommends review. It does not make the final legal or compliance decision.


BigQuery Transactions
Clean & Validate
Engineer Behaviour Features
Transparent Risk Score
Explain & Investigate
Alerts + Customer Summary

1. AML transaction monitoring

Simple meaning	A system that watches financial transactions and highlights activity that may need investigation.
Abbreviation	AML = Anti-Money Laundering.
Why it matters	Criminal activity can be hidden inside ordinary-looking payments. Monitoring helps investigators focus on the riskiest activity first.
Project use	The notebook reads transaction data, creates behavioural indicators, calculates a risk score and writes alert tables for investigators.
What can go wrong	Too many alerts waste investigator time; too few alerts can miss suspicious activity. A score may also be misunderstood as proof of crime.
How to validate / measure	Track alert volume, alert rate, precision, recall, false positives, false negatives, investigator disposition and time saved.

Professional Python implementation

```
# Count all transactions so the monitoring population is known.
total_transactions = len(df)

# Count transactions marked as alerts by the configured threshold.
total_alerts = int(df["is_alert"].sum())

# Calculate the percentage of transactions sent for review.
alert_rate = total_alerts / max(total_transactions, 1)

# Print the monitoring coverage for operational review.
print("Transactions:", total_transactions)
print("Alerts:", total_alerts)
print("Alert rate:", round(alert_rate, 4))
```

2. BigQuery data source and cloud analytics

Simple meaning	BigQuery is a cloud data warehouse that stores and analyses very large tables using SQL.
Abbreviation	SQL = Structured Query Language.
Why it matters	AML datasets can contain millions of rows. A cloud warehouse can scan, filter and aggregate them without loading everything into a laptop.
Project use	The project connects to a Google Cloud project, reads the source table and later writes alert and customer-risk tables back to BigQuery.
What can go wrong	Wrong project IDs, permissions, table names or unlimited queries can cause failures, cost overruns or exposure of sensitive data.
How to validate / measure	Confirm project and table IDs, inspect row counts and schema, use query limits during development, and review bytes processed.

Professional Python implementation

```
# Import the official BigQuery client library.
from google.cloud import bigquery

# Create a client connected to the approved Google Cloud project.
client = bigquery.Client(project=PROJECT_ID)

# Build a fully qualified table reference to avoid querying the wrong table.
table_ref = f"{PROJECT_ID}.{DATASET_ID}.{SOURCE_TABLE_ID}"

# Retrieve table metadata before downloading data.
table = client.get_table(table_ref)

# Print the table size so the analyst can plan a safe query.
print("Rows reported by BigQuery:", table.num_rows)
```

3. Authentication and least-privilege access

Simple meaning	Authentication proves who the user is. Authorisation decides what that user is allowed to do.
Abbreviation	IAM = Identity and Access Management.
Why it matters	Transaction data is sensitive. Only approved users and service accounts should read or write it.
Project use	The notebook authenticates the Colab user and creates a BigQuery client for the configured project.
What can go wrong	Overly broad access can expose data or allow accidental table deletion. Missing access blocks the pipeline.
How to validate / measure	Test read and write permissions separately, review IAM roles, inspect audit logs and remove unused credentials.

Professional Python implementation

```
# Import Colab authentication support.
from google.colab import auth

# Ask the current user to authenticate with Google.
auth.authenticate_user()

# Create a project-scoped BigQuery client after authentication succeeds.
client = bigquery.Client(project=PROJECT_ID)

# Display the connected project to prevent silent use of the wrong environment.
print("Connected project:", client.project)
```

4. Schema inspection and data contracts

Simple meaning	A schema describes each column name and data type. A data contract defines what the pipeline expects.
Abbreviation	ID = Identifier.
Why it matters	Code may silently produce wrong results when a date arrives as text or an amount arrives in the wrong column.
Project use	The notebook prints the BigQuery schema and searches for expected timestamp, bank and account columns.
What can go wrong	Renamed or duplicated account columns can mix sender and receiver information. Wrong types can break calculations.
How to validate / measure	Compare the actual schema with required fields, test data types, count missing required columns and stop the job when the contract fails.

Professional Python implementation

```
# Define the columns required for safe transaction analysis.
required_columns = {
    "transaction_timestamp",
    "from_bank",
    "from_account",
    "to_bank",
    "to_account",
    "amount_paid",
    "payment_currency",
}

# Find required columns that are absent from the dataframe.
missing_columns = sorted(required_columns - set(df.columns))

# Stop immediately when the data contract is broken.
if missing_columns:
    raise ValueError(f"Missing required columns: {missing_columns}")
```

5. Column-name standardisation

Simple meaning	Standardisation converts inconsistent names into one predictable format, such as lowercase snake_case.
Abbreviation	snake_case = words separated with underscores.
Why it matters	`From Bank`, `from-bank` and `FROM_BANK` should not be treated as different ideas.
Project use	The notebook removes special characters, replaces spaces with underscores and converts names to lowercase.
What can go wrong	Two original columns can become the same standardised name, or code can select the wrong automatically renamed account column.
How to validate / measure	Check for duplicate names after standardisation and compare mapped sender/receiver columns with source samples.

Professional Python implementation

```
# Import regular-expression support for controlled text replacement.
import re

# Define one reusable function for consistent column naming.
def normalise_column_name(name: str) -> str:
    # Convert any input label to text and remove outer spaces.
    name = str(name).strip()
    # Replace groups of non-alphanumeric characters with one underscore.
    name = re.sub(r"[^0-9a-zA-Z]+", "_", name)
    # Collapse repeated underscores and remove underscores at both ends.
    name = re.sub(r"_+", "_", name).strip("_")
    # Return lowercase snake_case for predictable downstream references.
    return name.lower()

# Apply the function to every dataframe column.
df.columns = [normalise_column_name(column) for column in df.columns]

# Detect collisions caused by standardisation.
assert not df.columns.duplicated().any(), "Duplicate column names detected."
```

6. Data cleaning and validation

Simple meaning	Cleaning converts raw values into usable formats. Validation checks whether those values make sense.
Abbreviation	NaN = Not a Number; UTC = Coordinated Universal Time.
Why it matters	Risk features are only reliable when timestamps, numbers, currencies and account IDs are correctly prepared.
Project use	The notebook converts dates and amounts, fills missing text safely, creates entity keys and checks transaction integrity.
What can go wrong	Invalid dates may become missing, negative amounts may distort risk, and blank account IDs may merge unrelated customers.
How to validate / measure	Measure missingness, invalid timestamps, negative or zero amounts, duplicate transaction IDs and impossible currency values.

Professional Python implementation

```
# Convert the transaction time to a timezone-aware datetime.
df["transaction_timestamp"] = pd.to_datetime(
    df["transaction_timestamp"],
    errors="coerce",
    utc=True,
)

# Convert payment amount to numeric and turn invalid text into missing values.
df["amount_paid"] = pd.to_numeric(df["amount_paid"], errors="coerce")

# Remove rows that cannot support time-based or amount-based analysis.
df = df.dropna(subset=["transaction_timestamp", "amount_paid"]).copy()

# Keep only economically meaningful positive payment amounts.
df = df.loc[df["amount_paid"] > 0].copy()

# Confirm the cleaned data still contains records.
assert len(df) > 0, "No valid transactions remain after cleaning."
```

7. Data-quality profiling

Simple meaning	Profiling creates a quick health report for every column.
Abbreviation	DQ = Data Quality.
Why it matters	Before modelling risk, the analyst must understand missing values, distinct values and data types.
Project use	The notebook builds a table with column name, type, missing count, missing percentage and unique count.
What can go wrong	A pipeline may appear successful while key columns are almost empty or contain one repeated value.
How to validate / measure	Use missing percentage, uniqueness, duplicate counts, range checks, category frequency and trend comparisons with previous runs.

Professional Python implementation

```
# Build one row of quality statistics for every dataframe column.
quality_summary = pd.DataFrame({
    # Store the column names being assessed.
    "column": df.columns,
    # Record the current pandas data type.
    "dtype": [str(df[column].dtype) for column in df.columns],
    # Count missing values in each column.
    "missing_count": [int(df[column].isna().sum()) for column in df.columns],
    # Calculate the missing-value percentage for easier comparison.
    "missing_pct": [
        round(float(df[column].isna().mean() * 100), 2)
        for column in df.columns
    ],
    # Count distinct values, including missing as a possible state.
    "unique_count": [
        int(df[column].nunique(dropna=False))
        for column in df.columns
    ],
})

# Sort the report so the most incomplete columns appear first.
quality_summary = quality_summary.sort_values(
    "missing_pct",
    ascending=False,
)
```

8. Entity keys and transaction identifiers

Simple meaning	An entity key combines fields to represent one sender or receiver consistently.
Abbreviation	ID = Identifier.
Why it matters	The same account number can exist at different banks, so account alone may not be unique.
Project use	The notebook combines bank and account to create sender_entity and receiver_entity, then creates a transaction_id.
What can go wrong	Weak keys can merge different customers, split one customer into many identities or double-count transactions.
How to validate / measure	Test uniqueness, inspect collisions, compare with master customer data and reconcile counts with the source system.

Professional Python implementation

```
# Build a sender key from both bank and account to reduce collisions.
df["sender_entity"] = (
    df["from_bank"].astype(str).str.strip()
    + "::"
    + df["from_account"].astype(str).str.strip()
)

# Build the equivalent receiver key.
df["receiver_entity"] = (
    df["to_bank"].astype(str).str.strip()
    + "::"
    + df["to_account"].astype(str).str.strip()
)

# Create a deterministic transaction ID when no trusted source ID exists.
df["transaction_id"] = (
    df.index.astype(str)
    + "-"
    + df["transaction_timestamp"].astype(str)
)

# Verify that every generated transaction ID is unique.
assert df["transaction_id"].is_unique
```

9. Behavioural feature engineering

Simple meaning	Feature engineering turns raw transactions into measurements that describe behaviour.
Abbreviation	Feature = a measurable input used by a rule or model.
Why it matters	A single amount may look normal, but repeated transfers, many receivers or unusual timing may reveal risk.
Project use	The notebook creates daily counts, daily amounts, unique receivers, receiver sender counts, time gaps and deviation measures.
What can go wrong	Data leakage, incorrect grouping, unsorted timestamps or future information can make performance look better than reality.
How to validate / measure	Recalculate features on a small hand-checked sample, test time ordering, compare group totals and ensure each feature uses only information available at that time.

Professional Python implementation

```
# Sort transactions before calculating time-dependent behaviour.
df = df.sort_values(
    ["sender_entity", "transaction_timestamp"]
).copy()

# Extract the calendar day used for same-day aggregations.
df["transaction_date"] = df["transaction_timestamp"].dt.date

# Count how many transactions each sender made on each day.
df["sender_daily_txn_count"] = df.groupby(
    ["sender_entity", "transaction_date"]
)["transaction_id"].transform("count")

# Sum the sender's total payment amount for that day.
df["sender_daily_amount"] = df.groupby(
    ["sender_entity", "transaction_date"]
)["transaction_amount"].transform("sum")

# Count how many different receivers the sender paid that day.
df["sender_daily_unique_receivers"] = df.groupby(
    ["sender_entity", "transaction_date"]
)["receiver_entity"].transform("nunique")
```

10. Velocity and rapid-repeat transfers

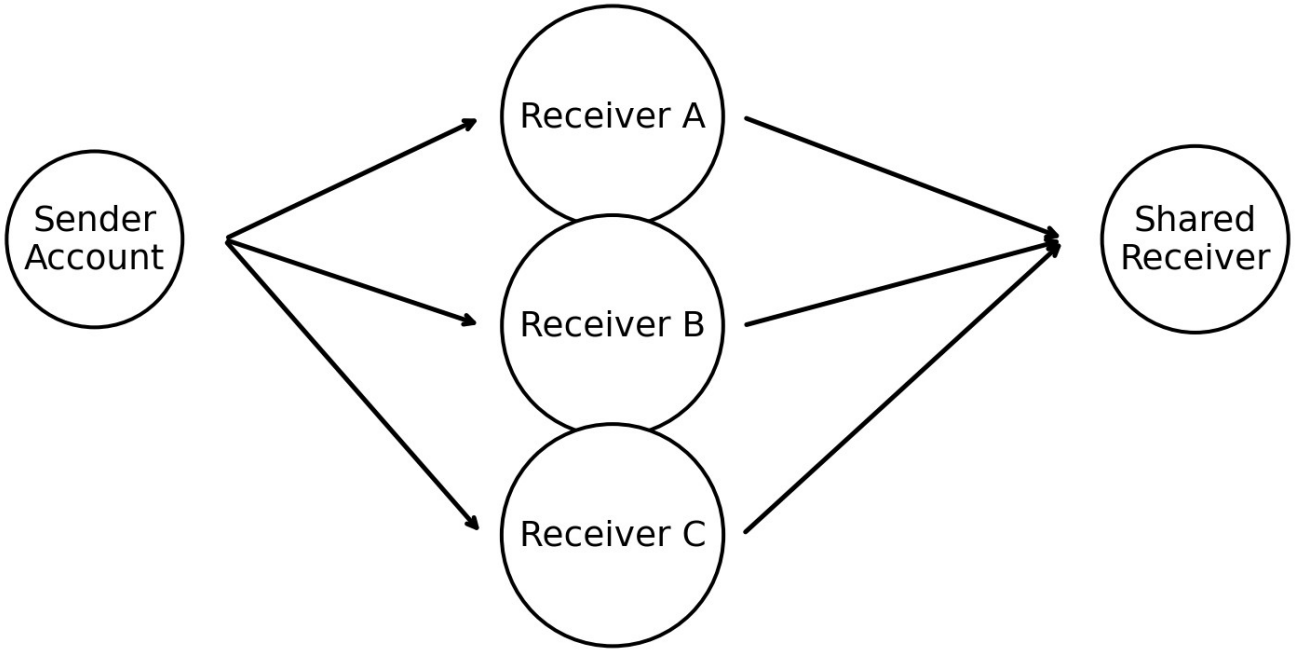
Simple meaning	Velocity measures how quickly and how often money moves.
Abbreviation	Txn = Transaction.
Why it matters	Many transactions in a short period can indicate automation, layering or deliberate splitting.
Project use	The notebook compares each sender transaction with the previous timestamp and marks another transfer within ten minutes.
What can go wrong	Unsorted data, mixed time zones or duplicated timestamps can create false rapid-transfer flags.
How to validate / measure	Inspect time gaps, compare flagged sequences with raw timestamps and measure alert precision for different time windows.

Professional Python implementation

```
# Find the previous transaction time for each sender.
df["previous_sender_timestamp"] = df.groupby(
    "sender_entity"
)["transaction_timestamp"].shift(1)

# Calculate elapsed minutes since the sender's previous transaction.
df["minutes_since_previous_sender_txn"] = (
    df["transaction_timestamp"]
    - df["previous_sender_timestamp"]
).dt.total_seconds() / 60.0

# Flag a repeated transfer occurring within ten minutes.
df["rapid_repeat_transfer"] = (
    df["minutes_since_previous_sender_txn"]
    .between(0, 10, inclusive="both")
    .fillna(False)
    .astype(int)
)
```

11. Network and counterparty behaviour

Simple meaning	Network analysis studies who sends money to whom and how many connections each party has.
Abbreviation	Fan-out = one sender pays many receivers; fan-in = one receiver collects from many senders.
Why it matters	Money-laundering behaviour may be visible in the relationship pattern rather than in one transaction.
Project use	The notebook counts a sender's unique receivers and a receiver's unique senders to capture fan-out and fan-in.
What can go wrong	Shared business accounts, payment processors and payroll accounts may look suspicious without customer context.
How to validate / measure	Compare with customer type, peer groups and known business purpose; inspect network diagrams and investigator outcomes.



Network features reveal fan-out, fan-in and linked activity patterns.

Professional Python implementation

```
# Count the number of distinct senders linked to each receiver.
df["receiver_unique_senders"] = df.groupby(
    "receiver_entity"
)["sender_entity"].transform("nunique")

# Flag receivers collecting from many different senders.
df["high_fan_in_receiver"] = (
    df["receiver_unique_senders"] >= 5
).astype(int)

# Flag senders paying many different receivers in one day.
df["high_fan_out_sender"] = (
    df["sender_daily_unique_receivers"] >= 4
).astype(int)
```

12. Statistical deviation and z-score

Simple meaning	A deviation feature asks whether a transaction is unusual for that sender.
Abbreviation	Z-score = number of standard deviations a value is above or below the mean.
Why it matters	A ₹5 lakh transfer may be normal for one business and highly unusual for another customer.
Project use	The notebook compares each transaction amount with the sender's historical mean and standard deviation.
What can go wrong	New customers have little history, outliers distort the mean, and zero standard deviation causes division problems.
How to validate / measure	Check distribution by customer, use minimum-history rules, compare robust alternatives such as median and MAD, and inspect extreme values.

Professional Python implementation

```
# Calculate each sender's average historical transaction amount.
sender_mean = df.groupby("sender_entity")[
    "transaction_amount"
].transform("mean")

# Calculate each sender's transaction-amount standard deviation.
sender_std = df.groupby("sender_entity")[
    "transaction_amount"
].transform("std")

# Avoid division by zero for senders with no variation.
safe_std = sender_std.replace(0, np.nan)

# Measure how many standard deviations each transaction is from the sender mean.
df["amount_zscore"] = (
    (df["transaction_amount"] - sender_mean) / safe_std
).fillna(0.0)

# Flag strongly unusual positive deviations.
df["high_amount_deviation"] = (
    df["amount_zscore"] >= 3.0
).astype(int)
```

13. Pattern indicators: structuring, round amounts and unusual hours

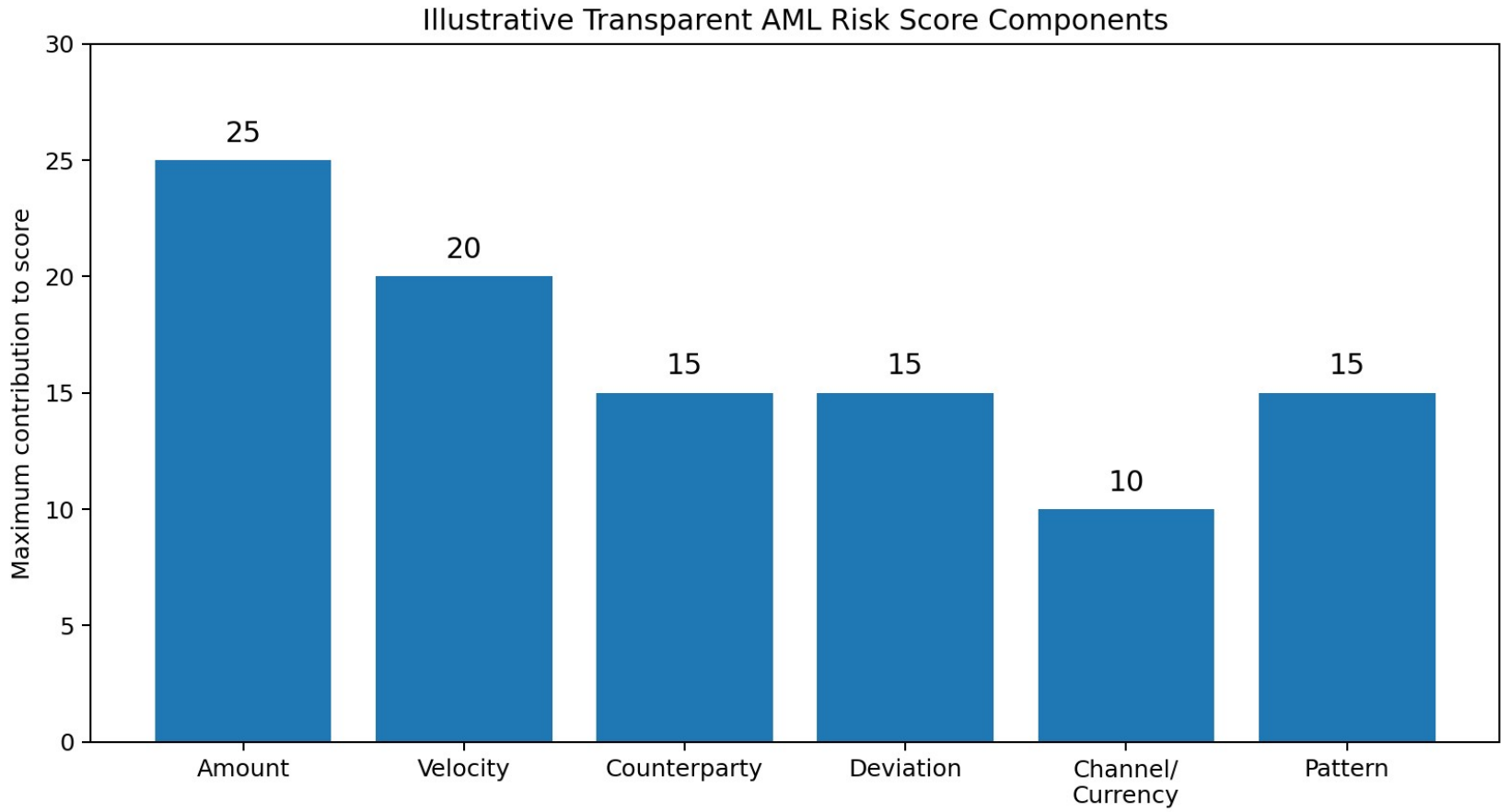
Simple meaning	Pattern indicators are understandable rules that point to known suspicious behaviours.
Abbreviation	Structuring = splitting money into several smaller transfers to avoid attention.
Why it matters	Transparent rules make it easier for investigators to understand the alert.
Project use	The notebook flags multiple same-day transfers, large round amounts, cash-like formats, self-transfers and unusual hours.
What can go wrong	Fixed thresholds may be too strict for one market and too loose for another. Legitimate customer habits can trigger alerts.
How to validate / measure	Back-test each rule, review hit rates by customer segment, tune thresholds and require investigator feedback.

Professional Python implementation

```
# Flag transactions made during low-activity overnight hours.
df["is_unusual_hour"] = (
    df["transaction_timestamp"].dt.hour.isin([0, 1, 2, 3, 4, 5])
).astype(int)

# Flag large values that are exact multiples of 1,000.
df["is_round_amount"] = (
    (df["transaction_amount"] >= 10000)
    & (df["transaction_amount"] % 1000 == 0)
).astype(int)

# Flag possible structuring when a sender makes several same-day payments.
df["potential_structuring"] = (
    (df["sender_daily_txn_count"] >= 3)
    & (df["sender_daily_amount"] >= 10000)
).astype(int)

# Flag transactions where sender and receiver are the same entity.
df["self_transfer"] = (
    df["sender_entity"] == df["receiver_entity"]
).astype(int)
```

14. Transparent weighted risk score

Simple meaning	A weighted score adds several risk components, with more important signals contributing more points.
Abbreviation	Score range in this project: 0 to 100.
Why it matters	Investigators can see which behaviours increased risk instead of trusting an unexplained model output.
Project use	Amount, velocity, counterparty, deviation, channel/currency and pattern components are combined and clipped to 100.
What can go wrong	Poor weights can over-alert one behaviour, hide another or create a false sense of mathematical certainty.
How to validate / measure	Compare score bands with known labels and investigator outcomes, perform sensitivity tests and monitor score distribution over time.


Illustrative Transparent AML Risk Score Components
Maximum contribution to score
0
5
10
15
20
25
30
25
20
15
15
10
15
Amount
Velocity
Counterparty
Deviation
Channel/
Currency
Pattern

Professional Python implementation

```
# Convert transaction amount into a relative 0-to-1 rank.
amount_rank = df["transaction_amount"].rank(
    pct=True,
    method="average",
)

# Give high-value transactions up to 25 risk points.
df["amount_risk_component"] = 25 * amount_rank

# Give transaction velocity up to 20 risk points.
df["velocity_risk_component"] = (
    10 * (df["sender_daily_txn_count"] >= 3).astype(int)
    + 10 * df["rapid_repeat_transfer"]
)

# Add the transparent components and cap the final score at 100.
df["risk_score"] = (
    df["amount_risk_component"]
    + df["velocity_risk_component"]
    + df["counterparty_risk_component"]
    + df["behaviour_deviation_component"]
    + df["channel_currency_component"]
    + df["pattern_component"]
).clip(lower=0, upper=100)
```

15. Thresholds, alerts and risk levels

Simple meaning	A threshold is the score boundary used to create an alert. Risk levels group scores into understandable bands.
Abbreviation	LOW, MEDIUM, HIGH and CRITICAL are operational severity labels.
Why it matters	The threshold controls workload and detection. A lower threshold creates more alerts; a higher threshold creates fewer.
Project use	The notebook uses a score of 60 or more as an alert and categorises the score into four levels.
What can go wrong	A threshold copied from development may overload investigators in production or miss important activity.
How to validate / measure	Review precision-recall trade-offs, alert capacity, false-negative cost, case-conversion rate and threshold stability.

Professional Python implementation

```
# Set the operational score boundary for creating an alert.
ALERT_THRESHOLD = 60.0

# Mark transactions meeting or exceeding the threshold.
df["is_alert"] = (
    df["risk_score"] >= ALERT_THRESHOLD
).astype(int)

# Convert numeric scores into human-readable risk bands.
df["risk_level"] = pd.cut(
    df["risk_score"],
    bins=[-np.inf, 30, 60, 80, np.inf],
    labels=["LOW", "MEDIUM", "HIGH", "CRITICAL"],
    right=False,
).astype(str)
```

16. Explainability and risk reasons

Simple meaning	Explainability tells the investigator which facts caused the score.
Abbreviation	XAI = Explainable Artificial Intelligence.
Why it matters	Compliance teams need defensible reasons, not just a number.
Project use	The notebook builds up to six plain-language reasons, such as large value, many receivers, rapid repetition or cross-currency transfer.
What can go wrong	Reasons may be technically true but misleading without customer context, or may not exactly match the score logic.
How to validate / measure	Unit-test every reason, reconcile reason text with component values and ask investigators whether explanations are actionable.

Professional Python implementation

```
# Start an empty explanation list for one transaction.
reasons = []

# Add a reason when the transaction amount is unusually high.
if row["amount_zscore"] >= 3:
    reasons.append(
        "Transaction amount is far above the sender's normal level."
    )

# Add a reason when the sender paid many receivers on the same day.
if row["sender_daily_unique_receivers"] >= 4:
    reasons.append(
        "Sender paid several different receivers on the same day."
    )

# Provide a safe fallback when risk comes from several moderate signals.
if not reasons:
    reasons.append(
        "Risk arises from the combined transaction and behavioural profile."
    )
```

17. Investigation assistant and human-in-the-loop review

Simple meaning	The assistant turns analytics into practical review steps and a draft narrative.
Abbreviation	HITL = Human in the Loop.
Why it matters	Investigators need context, recommended checks and consistent documentation, but a person must make the final decision.
Project use	The notebook recommends document checks, customer-behaviour comparison, network review and escalation for high-risk cases.
What can go wrong	Generated narratives can sound certain, contain incorrect context or encourage automatic decisions.
How to validate / measure	Require human approval, sample case quality, compare narrative facts with source columns and record investigator edits.

Professional Python implementation

```
# Define baseline review actions that apply to every alert.
actions = [
    "Verify the transaction purpose and supporting documentation.",
    "Compare activity with the customer's expected account behaviour.",
]

# Add a targeted action when possible structuring is present.
if row["potential_structuring"] == 1:
    actions.append(
        "Review same-day transfers for linked or deliberately split payments."
    )

# Escalate severe cases only after contextual evidence is reviewed.
if row["risk_level"] in {"HIGH", "CRITICAL"}:
    actions.append(
        "Escalate for enhanced review if evidence does not explain the activity."
    )
```

18. Transaction alert table

Simple meaning	An alert table contains only transactions selected for investigation, together with evidence and actions.
Abbreviation	Case queue = the ordered list of alerts awaiting review.
Why it matters	Investigators need a compact, prioritised dataset instead of the full transaction population.
Project use	The notebook selects alert columns, filters is_alert = 1 and sorts by risk score from highest to lowest.
What can go wrong	Missing evidence columns prevent investigation; duplicated alerts create wasted work; sensitive columns may be overexposed.
How to validate / measure	Reconcile alert counts, verify sort order, test required fields, check duplicates and confirm access controls.

Professional Python implementation

```
# Select only records that crossed the alert threshold.
alerts = df.loc[
    df["is_alert"] == 1,
    alert_columns,
].copy()

# Sort the queue so the highest-risk cases are reviewed first.
alerts = alerts.sort_values(
    "risk_score",
    ascending=False,
).reset_index(drop=True)

# Verify that every row in the queue is truly an alert.
assert alerts["is_alert"].eq(1).all()
```

19. Customer-level risk aggregation

Simple meaning	Aggregation combines many transactions into one summary row for each sender account.
Abbreviation	KYC = Know Your Customer.
Why it matters	A customer may look risky because of repeated behaviour even when no single transaction appears extreme.
Project use	The notebook calculates totals, averages, maximums, unique receivers, alert counts, alert rate and maximum risk score.
What can go wrong	Aggregation can hide timing, mix unrelated entities or label a customer using one erroneous transaction.
How to validate / measure	Reconcile account totals, drill down from customer to transactions, test key quality and compare with KYC/customer type.

Professional Python implementation

```
# Group transaction-level data into one row per sender entity.
customer_risk = df.groupby("sender_entity").agg(
    # Count all transactions initiated by the sender.
    total_transactions=("transaction_id", "count"),
    # Sum the sender's transaction amount.
    total_amount=("transaction_amount", "sum"),
    # Count how many alerts were generated for the sender.
    alert_count=("is_alert", "sum"),
    # Keep the highest transaction-level risk score.
    maximum_risk_score=("risk_score", "max"),
    # Count distinct counterparties paid by the sender.
    unique_receivers=("receiver_entity", "nunique"),
).reset_index()

# Calculate the proportion of the sender's transactions that became alerts.
customer_risk["alert_rate"] = (
    customer_risk["alert_count"]
    / customer_risk["total_transactions"].replace(0, np.nan)
).fillna(0.0)
```

20. Model validation with labelled data

Simple meaning	Validation compares alerts with known outcomes to understand how well the system detects labelled laundering cases.
Abbreviation	TP = True Positive; FP = False Positive; FN = False Negative; TN = True Negative.
Why it matters	A score is not trustworthy until its behaviour is measured on relevant data.
Project use	When the IBM laundering label is available, the notebook calculates a confusion matrix, classification report, ROC-AUC and average precision.
What can go wrong	Labels may be incomplete, simulated or not representative of current production behaviour. Accuracy can be misleading for rare crime events.
How to validate / measure	Prioritise precision, recall, F1, PR-AUC, alert workload, stability and investigator outcomes; validate on separate time periods.

Professional Python implementation

```
# Store the known laundering label as the validation target.
y_true = df["is_laundering"].astype(int)

# Store the pipeline's binary alert decision.
y_pred = df["is_alert"].astype(int)

# Convert the 0-to-100 score into a 0-to-1 ranking value.
y_score = df["risk_score"] / 100.0

# Print the confusion matrix to inspect TP, FP, FN and TN counts.
print(confusion_matrix(y_true, y_pred))

# Measure ranking quality for the rare positive class.
print("Average precision:", average_precision_score(y_true, y_score))
```

21. Precision, recall, F1 and threshold tuning

Simple meaning	Precision asks how many alerts were truly positive. Recall asks how many true positives were found. F1 balances both.
Abbreviation	F1 = harmonic mean of precision and recall; PR = Precision-Recall.
Why it matters	AML teams must balance missed risk against investigator workload.
Project use	The notebook calculates precision and recall at many thresholds, then identifies the threshold with the best F1 score.
What can go wrong	The mathematically best F1 threshold may not match regulatory risk appetite or available case capacity.
How to validate / measure	Review the full precision-recall curve, expected cases per day, cost of missed cases and performance by customer segment.

Professional Python implementation

```
# Calculate precision and recall across possible score thresholds.
precision, recall, thresholds = precision_recall_curve(
    y_true,
    y_score,
)

# Align threshold values with the precision and recall arrays.
threshold_table = pd.DataFrame({
    "threshold": np.r_[thresholds, np.nan],
    "precision": precision,
    "recall": recall,
})

# Calculate F1 while safely handling a zero denominator.
threshold_table["f1"] = (
    2
    * threshold_table["precision"]
    * threshold_table["recall"]
    / (
        threshold_table["precision"]
        + threshold_table["recall"]
    ).replace(0, np.nan)
)

# Select the row with the highest measured F1 value.
best_threshold_row = threshold_table.loc[
    threshold_table["f1"].idxmax()
]
```

22. Writing governed outputs back to BigQuery

Simple meaning	The pipeline saves approved results as managed cloud tables for dashboards and investigator queries.
Abbreviation	WRITE_TRUNCATE = replace the destination table with the new run.
Why it matters	Operational users need stable outputs that can be queried independently of the notebook.
Project use	The notebook uploads transaction alerts and customer summaries to two BigQuery tables.
What can go wrong	WRITE_TRUNCATE can erase history, schema drift can fail uploads and repeated runs may remove auditability.
How to validate / measure	Check destination row counts, schema, load-job status, run timestamps, lineage and retention strategy.

Professional Python implementation

```
# Configure the load to replace the previous table contents.
job_config = bigquery.LoadJobConfig(
    write_disposition="WRITE_TRUNCATE",
)

# Start the alert-table upload to BigQuery.
alert_load_job = client.load_table_from_dataframe(
    alerts_to_upload,
    OUTPUT_ALERT_TABLE,
    job_config=job_config,
)

# Wait for the job to finish so failures are not ignored.
alert_load_job.result()

# Confirm the destination table now contains the expected number of rows.
uploaded_table = client.get_table(OUTPUT_ALERT_TABLE)
assert uploaded_table.num_rows == len(alerts_to_upload)
```

23. Investigator SQL queries

Simple meaning	SQL queries let investigators retrieve the highest-risk alerts and repeated high-risk accounts.
Abbreviation	ORDER BY = sort results; LIMIT = return only a selected number of rows.
Why it matters	Operational teams need fast, repeatable ways to find cases requiring attention.
Project use	The notebook queries the alert table by descending risk score and filters customer summaries to HIGH or CRITICAL.
What can go wrong	Queries without filters can scan too much data, expose sensitive fields or produce inconsistent case lists.
How to validate / measure	Use approved views, parameterised filters, row-level security, query-cost monitoring and result reconciliation.

Professional Python implementation

```
# Define a query that returns the highest-risk transaction alerts.
top_alerts_sql = f'''
SELECT
  transaction_id,
  transaction_timestamp,
  from_bank,
  from_account,
  to_bank,
  to_account,
  transaction_amount,
  risk_score,
  risk_level,
  risk_reasons
FROM `{OUTPUT_ALERT_TABLE}`
ORDER BY risk_score DESC
LIMIT 100
'''

# Execute the approved query and load the result into pandas.
top_alerts = client.query(top_alerts_sql).to_dataframe()
```

Production readiness checklist

Data governance	Encrypt sensitive data, apply least privilege, mask account details where possible and retain audit logs.
Rule governance	Document every rule, owner, rationale, version, threshold and approval date.
Model risk	Validate on separate periods and customer segments; monitor drift and false-negative exposure.
Operations	Set alert-capacity limits, prioritisation rules, SLAs and escalation paths.
Human oversight	Require investigators to approve or reject alerts and record reasons.
Auditability	Store run ID, code version, data date, score components, explanations and final disposition.
Monitoring	Track data quality, alert rate, precision, recall, average case time and changes in score distribution.
Change management	Test updates in a non-production environment and use controlled deployment.

Interview revision: 12 questions with strong answer points

Interview question	Strong answer points
Why use a transparent score instead of only a black-box model?	AML decisions require explainability, auditability and investigator trust. A transparent score also makes threshold and rule governance easier.
What is the difference between transaction risk and customer risk?	Transaction risk evaluates one payment. Customer risk aggregates behaviour across many payments and counterparties.
Why sort by sender and timestamp before velocity features?	Time gaps and previous-transaction features are only correct when events are in chronological order for each sender.
Why can accuracy be misleading?	Laundering labels are rare, so a model can achieve high accuracy by predicting almost everything as normal.
What does precision mean operationally?	Of all alerts sent to investigators, precision is the proportion that match the positive label.
What does recall mean operationally?	Of all labelled laundering transactions, recall is the proportion detected by the system.
Why use BigQuery?	It supports scalable SQL, governed cloud storage, large-table aggregation and shared operational outputs.
What is data leakage?	Using future or outcome information that would not be available at the time of scoring.
Why combine bank and account in an entity key?	Account numbers may repeat across banks; the combination reduces identity collisions.
Why is a threshold a business decision?	It controls both missed-risk exposure and investigation workload.

What is human-in-the-loop?	Analytics recommends and explains; a trained investigator validates context and decides the final disposition.
What would you improve for production?	Add peer groups, time-windowed historical features, feature store, drift monitoring, model/rule versioning, case feedback and stronger privacy controls.

Final candidate summary

What the project does	Converts raw transaction data into prioritised AML alerts and account-level risk summaries.
What makes it professional	Uses governed cloud data, data validation, behavioural features, transparent components, measurable thresholds and human review.
Most important interview message	The score is a decision-support mechanism, not proof of laundering.
Best improvement statement	Production readiness depends on data governance, back-testing, investigator feedback, monitoring and controlled threshold changes.

The Talent Grid | Learn the concept, explain the business value, validate the evidence.